



SmartAttend

AI Face-Recognition Attendance

USER MANUAL

Installation • Face Registration • Teacher Attendance & Exports

Version 3.0 | Android & iOS

Contents

1. Introduction
2. APK Installation & Permissions Guide
3. First-Time Setup & Login
4. Face Registration — Lighting Best Practices
5. App Roles at a Glance
6. Teacher Guide — Session Activation & Exports
7. Student Guide
8. Settings Reference
9. Troubleshooting
10. Support

1. Introduction

SmartAttend is a mobile attendance system that uses face recognition to take classroom attendance from a single group photo. A teacher captures or uploads a photo of the class, the app detects and matches faces against enrolled students, and attendance is recorded automatically — with a manual review step before anything is saved.

This manual covers three roles:

- Admin — manages universities, teacher accounts, and system-wide reports.
- Teacher — creates classes and sections, enrolls students, takes attendance, and exports records.
- Student — completes their profile photo, browses/enrolls in classes, views attendance history, and submits leave requests.

SCOPE OF THIS MANUAL

This guide focuses on installing the app, registering a usable face photo, and the teacher workflow for activating an attendance session and exporting data. Admin-only screens are mentioned only where relevant.

2. APK Installation & Permissions Guide

2.1 Installing the Android APK

1. On your Android device, open the SmartAttend APK file (received via link, email, or file transfer).
2. If this is the first APK you're installing from this source, Android will show "Install blocked" or "For your security, your phone is not allowed to install unknown apps from this source." Tap Settings on that prompt.
3. Enable "Allow from this source" (the exact wording depends on Android version: Settings → Apps → Special app access → Install unknown apps → select the app you used to open the file, e.g. Files or Chrome → toggle Allow from this source).
4. Return to the APK file and tap Install.
5. Once installation finishes, tap Open, or find the SmartAttend icon on your home screen / app drawer.

iOS

iOS builds are distributed via TestFlight or an internal Ad Hoc/Enterprise link provided by your institution, since sideloading .ipa files works differently than Android APKs. Follow the install link you were given and trust the developer profile under Settings → General → VPN & Device Management if prompted.

2.2 Permissions the App Requests

SmartAttend asks for the following permissions the first time each feature is used — not all at once at launch. You can always change a decision later from your device's app settings.

Permission	Platform	Why SmartAttend needs it
Camera	Android & iOS	Required to capture classroom photos for attendance and to take/update your profile photo used for face recognition.
Photos / Storage (Read & Write)	Android & iOS	Lets you choose an existing photo from your gallery instead of the camera, and lets the app save exported files (attendance exports, data export) to your device.

Permission	Platform	Why SmartAttend needs it
Microphone	Android	Reserved for the camera module on some Android devices that bundle audio capture with the camera API; SmartAttend does not record audio.
Notifications	Android & iOS	Needed to deliver push notifications (enrollment updates, attendance results, leave decisions) to your device. You can turn this on/off anytime in Settings → Push Notifications.

IF YOU DENY A PERMISSION

You can still use most of the app, but any screen that needs the denied permission (e.g. Camera for Take Attendance) will show an in-app prompt explaining what to enable and where. Re-grant permissions from your phone's Settings → Apps → SmartAttend → Permissions.

3. First-Time Setup & Login

3.1 Creating an account

- Teachers and Admins register/sign in with an email address and password.
- Students can register with either an email or a Student ID — a teacher may also create a student's account directly, in which case the student receives a temporary password and is required to set a new one on first login.

3.2 Sign-in gates, in order

- Forced password change — if your account was created by a teacher/admin with a temporary password, you must set a new password before continuing.
- Complete profile (students only) — students must upload a clear profile photo before reaching the dashboard, since this photo is what attendance recognition is matched against. See Section 4 for how to get a good result on the first try.
- Dashboard — once the above are satisfied, you land on your role's home dashboard.

4. Face Registration — Lighting Best Practices

Your profile photo is the single reference the recognition engine uses to identify you in every future class photo. A poor-quality reference photo is the single biggest cause of a student showing up as "Unrecognized" on attendance day — so it's worth getting right once.

4.1 What the app checks automatically

When you upload or retake your profile photo, SmartAttend runs an automatic quality check before accepting it:

- Exactly one face must be detected — photos with zero faces (too dark, blurry, face turned away) or more than one face (group selfies) are rejected.
- If the check fails, you'll see a clear in-app message explaining why, and you can immediately retake or re-pick the photo.

PASSING THE AUTOMATIC CHECK IS THE MINIMUM BAR

A photo can pass this check and still recognize poorly in a real classroom photo (e.g. if it was taken in near-darkness with heavy shadow). Follow the guidance below for a photo that performs well, not just one that gets accepted.

4.2 Lighting best practices

Do

- Face a window or light source — position yourself so light falls evenly on your face from the front, not from behind you.
- Use soft, diffused light — daylight through a window or standard indoor room lighting works better than a single harsh lamp.
- Keep the light source in front of or beside you, roughly at eye level.
- Make sure both eyes, eyebrows, nose, and mouth are clearly visible and well lit, with no strong shadow across half the face.
- Remove sunglasses, masks, or anything covering the lower or upper half of your face.
- Use a plain, evenly lit background so the face-detector isn't confused by clutter.
- Hold the phone at roughly arm's length, at eye level, facing the camera directly (not a steep up/down angle).

Avoid

- Backlighting — never take the photo with a bright window or light directly behind you; this silhouettes your face and hides the detail the engine needs.
- Very dim or nighttime rooms with only a single overhead light — this creates heavy shadows under the eyes and jaw.
- Mixed colored lighting (e.g. strong colored LED lamps) that shifts your natural skin tone.
- Hats, hoods, or hair covering the forehead or eyebrows.
- Heavy filters, beauty-mode smoothing, or extreme close-ups that crop out part of the face.

4.3 Quick checklist before you tap capture

Check	Look for
Light direction	Light is in front of you, not behind
Shadows	No dark shadow across one side of the face
Face coverage	Eyes, nose, mouth, eyebrows all visible
Framing	One face only, centered, eye-level angle
Background	Plain wall or simple background
Focus	Image is sharp, not blurry or over-compressed

RE-UPLOADING A PHOTO

You can update your profile photo anytime from Settings → Edit Profile. Re-uploading immediately replaces the face data used for recognition — do this if your appearance changes significantly (new glasses, facial hair, etc.) or if you were originally showing up as Unrecognized often.

5. App Roles at a Glance

Role	What they can do
Admin	Manage universities and teacher accounts, view system-wide reports and analytics, export CSV reports, review recent activity logs.
Teacher	Create/edit classes and sections, enroll students (manually, by CSV, or with a photo), take and review attendance, manage enrollment and leave requests, view class rosters and student stats, export personal data.
Student	Complete profile with a face photo, browse and request enrollment in classes, view personal attendance history and stats, submit leave requests, export personal data.

6. Teacher Guide — Session Activation & Exports

6.1 Before taking attendance

- Create your class and at least one section (Classes → + New Class), setting the schedule days and start/end time for each section.
- Enroll students into the class — manually, by pasting a CSV/text list, or with a photo (Class → Students → add-student icon). Each enrolled student needs face data on file (their own profile photo, or one captured during enrollment) for recognition to work.

6.2 Activating an attendance session

An attendance "session" is one classroom photo processed and saved for a specific section, for a specific date. SmartAttend allows exactly one saved session per section per day.

1. From the dashboard, tap the center camera button, or open a class and choose Take Attendance.
2. Select the section you're taking attendance for.
3. If a session already exists for that section today, you'll be offered: Update Records (edit who's present/absent), Capture New Photo (replace today's photo and redo recognition), or Delete & Retake.
4. Capture a classroom photo with the in-app camera, or upload an existing photo from your gallery. For best recognition results, take the photo with the whole class facing the camera in reasonably even lighting — the same lighting principles from Section 4 apply at the group level.
5. The app processes the photo through several stages — detecting faces, checking for masked/occluded faces, matching against your enrolled roster, and computing Present/Absent — while showing the captured photo with a live progress indicator.

6.3 Reviewing results

After processing, the Attendance Review screen shows the annotated class photo plus a strip of every detected face, each labeled Present, Absent, or Unrecognized:

- Tap a recognized student's face (or their row in the roster list below) to manually flip their status between Present and Absent.
 - An Unrecognized face means the detector found a person but couldn't confidently match them to anyone on your roster (or they aren't enrolled yet). Tap that face to open the "Add Unrecognized Student" dialog.
1. In the dialog, enter the student's full name and Student ID.

2. If that Student ID already has an account, they are simply enrolled into this class/section — their existing profile photo is kept.
3. If it's a new Student ID, a new account is created and the cropped face from this photo becomes their profile photo, so they're recognized automatically next time.
4. Tap Enroll & Mark Present to confirm.

STUDENTS MISSING FACE DATA

If enrolled students have no face data on file yet (e.g. they haven't uploaded a profile photo), the review screen flags their names so you know to ask them to complete their profile before the next session.

6.4 Saving, editing, and deleting a session

- Tap Save (or Save Changes in edit mode) once every student's status looks correct.
- To correct an already-saved session later the same day, reopen Take Attendance for that section and choose Update Records.
- To discard a session entirely and start over, choose Delete Today's Attendance from the review screen, then retake the photo.

6.5 Exporting attendance and account data

SmartAttend keeps two separate export paths:

- Class-level attendance history — reviewable per class/section under History, for day-to-day reference inside the app.
- Personal data export — from Settings → Data Export, available to every role including teachers.

Using Data Export

1. Open Settings → Data Export.
2. Choose a format: Excel (.xlsx) — a spreadsheet workbook with separate sheets for your Profile, Classes, and Notifications, ready to open in Excel, Google Sheets, or Numbers — or JSON, a structured file for developers/system integration.
3. Tap Export Data. The file is generated on your device and the share sheet opens automatically so you can save it, email it, or send it to another app.
4. If sharing isn't available on your device, the app shows you the exact file path where the export was saved instead.

WHAT'S INCLUDED

The export compiles real-time data linked to your signed-in account: your profile, your classes, and your notifications. It is a personal export, not a full class roster or attendance ledger — use the in-app History/Reports screens to review other students' attendance.

7. Student Guide

7.1 Getting enrolled

- Complete your profile photo on first login (see Section 4 for lighting tips) — this is what attendance recognition matches against.

- Browse Classes to find and request enrollment in a class, or wait to be enrolled directly by a teacher.
- Track the status of your enrollment/leave requests from the Notifications tab — tapping a notification opens the relevant screen.

7.2 Attendance & leave

- View your attendance history and per-class stats from History and Student Stats.
- Submit a leave request for a date you'll miss class; your teacher approves or rejects it, and you're notified either way.

8. Settings Reference

Setting	What it does
Push Notifications	Turns on/off push alerts for enrollment, attendance, and leave events on this device. Turning it on requests OS notification permission; turning it off stops further pushes to this device.
Dark Mode	Switches the app's color theme.
Auto-save Photos	When enabled, every classroom photo captured for attendance is also saved to your device's local storage in addition to being processed by the app.
Privacy Policy	Opens SmartAttend's privacy policy.
Data Export	Exports your profile, classes, and notifications as an Excel (.xlsx) or JSON file (see Section 6.5).
Change Password	Update your account password.

9. Troubleshooting

Issue	What to try
"Install blocked" on Android	Enable "Allow from this source" for the app you used to open the APK, per Section 2.1.
Camera/photos won't open	Check Settings → Apps → SmartAttend → Permissions on your device and re-enable Camera / Photos.
My profile photo keeps getting rejected	Make sure only one face is in frame, well lit and facing the camera — see Section 4.
I show up as Unrecognized in class photos	Re-upload your profile photo with better front-facing lighting (Section 4.3 checklist), and ask your teacher to add you from the Unrecognized tile if it happens again in a saved session.
Not receiving push notifications	Confirm the Push Notifications toggle is on in Settings, and that notifications are allowed for SmartAttend in your device's system settings.
Export file didn't share	If no share sheet appears, the app shows the on-device file path directly in a confirmation message — locate it with a file manager app.
"Couldn't reach the server"	Check your internet connection; the app needs connectivity for login, recognition, and syncing records.

10. Support

For account issues (locked out, wrong role, missing enrollment), contact your class teacher or your institution's SmartAttend administrator first — many issues (password resets, re-enrollment) can be resolved directly from within the app under Settings or Forgot Password.

For app bugs or feature requests, use your institution's designated support channel or IT helpdesk.